

# BiLSTM-based Language Identification in Code-Switched Social Media Text

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**Abstract**—Code-switching, the practice of alternating between languages within a conversation or even within a single sentence, is common in multilingual social media posts. Accurate word-level language identification (LID) in such code-switched text is crucial for downstream NLP tasks (e.g., translation, POS tagging), but it is challenging due to noise, informal spelling, and lexical overlap between languages. We propose a bidirectional Long Short-Term Memory (BiLSTM) network that integrates character-level and word-level representations to label each token as English, the other language (Hindi or Spanish), or Other. Experiments on benchmark English-Hindi and English-Spanish Twitter datasets demonstrate that our BiLSTM model substantially outperforms traditional baselines. Specifically, it achieves an overall accuracy of 96.8%, compared to 89.2% for an SVM and 91.5% for a CRF classifier. We present a detailed error analysis: noisy transliteration and ambiguous context at switch points are major sources of error. Our results show that deep contextual models can effectively capture the patterns of code-switching in noisy social text. The code and data will be made available for reproducibility.

**Index Terms**—Code-Switching, Language Identification, BiLSTM, Social Media Text, Multilingual NLP, Deep Learning

## I. INTRODUCTION

Code-switching refers to the use of multiple languages within the same utterance or discourse [1]. It is a widespread phenomenon in informal social media communication in multilingual communities (e.g., India, Latin America) [6]. For example, Hindi-English bilinguals often mix Hindi phrases with English words in a single tweet or sentence. While such mixed-language data is rich in sociolinguistic information, it poses challenges for NLP systems. In particular, standard monolingual tools (e.g., part-of-speech taggers, parsers) typically assume a single language and break down when faced with code-switched text. Word-level language identification (LID) is a critical first step in processing code-switched content, as it enables subsequent modules to apply the correct language model for each token [4].

However, token-level LID for code-switched social media text is inherently difficult. Text from Twitter or Facebook is often noisy and informal (slang, spelling variants, repetitive characters), and many tokens may be ambiguous or transliterated [7]. For example, Hindi words appear in Roman script, and English and Spanish share similar vocabulary due to Latin roots, leading to lexical overlap [2]. These factors make it hard for simple classifiers to distinguish languages. Recently, neural sequence models such as BiLSTM have shown promise in capturing context and subword patterns [5], [8]. Motivated by these advances, we develop a BiLSTM-based LID system tailored to code-switched social media text.

In this study, we scrutinize the efficacy of advanced neural architectures in addressing the challenges of code-switched language identification (LID). Our research agenda is centered around developing a deep contextual model that can effectively harness morphological and contextual cues to facilitate accurate LID. We introduce a novel hierarchical BiLSTM model, which encodes tokens at both character and word levels, thereby bridging the gap between token-level and sequence-level representations.

Our investigation is complemented by comparative evaluations on publicly available code-switched Twitter datasets for English-Hindi and English-Spanish languages. We engage in a systematic analysis of our proposed model against established baselines, namely Support Vector Machines (SVM) and Conditional Random Fields (CRF). Our results are presented in a quantitative manner through Tables and Figures, which provide a comprehensive overview of the model's performance.

To further elucidate the strengths and weaknesses of our approach, we conduct an exhaustive error analysis, highlighting common pitfalls that hinder accurate LID. Notably, our investigation reveals instances of noisy spelling and ambiguity at switch points, which underscore the importance of developing contextual models that can effectively capture these nuances.

The empirical evidence from our experiments unequivocally

demonstrates the superiority of our BiLSTM model over baselines, achieving an accuracy of 96.8. This finding highlights the value of adopting deep contextual modeling strategies in addressing the complexities of code-switched language identification.

## II. RELATED WORK

The recent surge in code-switched language identification research has been driven by the development of advanced neural models and large-scale datasets. These studies have highlighted the need for contextual information and global sentence structure in code-switching detection. Notably, Nguyen et al.'s [3] comprehensive datasets for Hindi-English code-switching from social media platforms have significantly contributed to this endeavor.

Moreover, Burchell et al.'s [1] findings on the challenges of code-switched language identification in large-scale multilingual settings underscore the importance of leveraging contextual information and sentence-level multi-label formulations. Their work demonstrates that traditional token-level approaches can be improved upon by incorporating global sentence structure.

Furthermore, Kargaran et al.'s [4] unsupervised method for code-switching identification using iterative masking strategies has shown promising results in low-resource scenarios, leveraging transformer architectures to achieve strong performance even without extensive labeled data. Similarly, De León et al.'s [2] probing framework reveals that pre-trained language models can effectively generalize to code-switched text when properly fine-tuned.

Udawatta et al.'s [5] exploration of prompt-based learning for code-mixed and code-switched text classification highlights the potential of carefully designed prompts in improving model performance without extensive architectural changes. Ghafouri et al.'s [7] PinLID dataset for Penglish language identification underscores the need for language-specific approaches in code-switching research.

Recent advances in transformer-based architectures have also enabled significant improvements in code-switched language identification, as demonstrated by Zhang et al.'s [8] proposal of a cross-lingual contextual embeddings approach. Chen et al.'s [9] investigation into the role of syntactic features in code-switched language identification highlights the importance of incorporating dependency parsing information for improved model performance.

The exploration of meta-learning for few-shot code-switched language identification by Kumar et al. [11] and the comprehensive survey of code-switched Arabic NLP by Hamed et al. [6] further underscore the evolving landscape of code-switching research, emphasizing neural approaches, multilingual capabilities, and robust evaluation frameworks.

These studies collectively demonstrate a shift towards more nuanced understanding of code-switching detection, highlighting the need for contextual information, global sentence structure, and language-specific approaches in this domain.

## III. METHODOLOGY

In our proposed language identification model, a hierarchical architecture leverages sequential representations to effectively capture code-switched text complexities. The model's configuration can be visualized through Figure ??, which illustrates the layer-by-layer processing of tokens. Each token within the input sequence is processed through two distinct sub-networks: one focused on word-level semantics and the other emphasizing character-level spelling information.

The word-level representation of each token is generated by a learnable word embedding, potentially initialized with pre-trained vectors to facilitate efficient fine-tuning. Conversely, a character-level BiLSTM is employed to encode the token's spelling, producing a fixed-dimensional vector that encapsulates its unique morphological characteristics. The output of this character-level sub-network serves as a crucial bridge between sequence-level and token-level representations.

The full token representation is subsequently formed by concatenating the word embedding with the character-level representation, thereby effectively integrating both semantic and spelling information into a unified token vector. This novel approach to sequential processing enables our model to robustly handle code-switched text, where linguistic boundaries often blur across words and characters.

These token vectors are fed into a second-level BiLSTM over the sentence, capturing context from both preceding and following tokens. This contextual BiLSTM has hidden size 100 in each direction. Finally, a softmax output layer predicts one of three labels for each token: LANG1 (e.g., English), LANG2 (Hindi or Spanish), or OTHER (non-language tokens such as named entities, emojis, etc.). We train the model end-to-end with cross-entropy loss and backpropagation through time. During training we use dropout (0.5) on the BiLSTM outputs to mitigate overfitting. We optimize with the Adam algorithm and early-stop on a held-out development set. All models (BiLSTM, SVM, CRF baselines) use the same preprocessing: converting to lowercase, tokenizing by whitespace and punctuation, and labeling each token with one language tag.

For comparison, we implement two baseline classifiers. The first is a linear Support Vector Machine (SVM) that classifies each token independently. We extract character n-gram features (lengths 1–5) and word n-grams (up to length 3) in a TF-IDF representation. We train the SVM in a one-vs-all fashion on the training set labels. The second baseline is a linear-chain Conditional Random Field (CRF) sequence model. The CRF uses similar features: for each token we include character n-grams (1–5), a boolean indicator for word shape (capitalization pattern), and context windows of size 2. The CRF outputs one label per token and benefits from sequential information via context features.

In summary, the BiLSTM model learns dense embeddings and contextual patterns automatically, while the SVM and CRF rely on hand-crafted n-gram features. All models are trained

on the same datasets and evaluated by per-token accuracy on held-out test sets.

#### IV. DATASET DESCRIPTION

We conduct experiments on code-switched Twitter datasets involving English mixed with either Hindi or Spanish. For English–Hindi, we use the publicly available ICON-2021 dataset [3], which contains Hindi-English tweets annotated with word-level language tags (Hindi, English, Other) from Facebook, Twitter and WhatsApp sources. This dataset contains approximately 15,000 tokens of annotated text with diverse code-switching patterns.

For English–Spanish, we use a recently collected Twitter dataset following the annotation framework proposed by De León et al. [2]. This corpus contains approximately 5,000 tweets with token-level language annotations, representing natural code-switching patterns in Spanish-English bilingual communities.

In both datasets, each token is tagged as the content of Language 1 (English), Language 2 (Hindi or Spanish), or Other (named entities, foreign fragments, emoticons, etc.). We combine the two corpora only in some multilingual experiments; otherwise we report results on each language pair separately.

Preprocessing involves basic normalization: tweets are lowercased, hashtags and user mentions are retained as tokens (labeled Other), and non-textual tokens (URLs, emoticons) are filtered out. We split each dataset into training (80%), development (10%), and test (10%) sets, ensuring that no tweet appears in more than one split. Both datasets exhibit the typical code-switching characteristics noted in recent research, including transliteration challenges and lexical borrowing.

#### V. EXPERIMENTS

We train the BiLSTM model and baselines on each code-switched dataset. Hyperparameters (e.g., SVM regularization, CRF L2 penalty) are tuned on the development set. For the BiLSTM, we fix a word embedding size of 100 and char embedding size of 50, and run for up to 30 epochs with early stopping. We report accuracy on the test set as the evaluation metric (percentage of tokens correctly labeled).

As baselines, we use the SVM and CRF classifiers described above. For the SVM we use a linear kernel with  $C = 1.0$ . For the CRF we use the CRF++ toolkit with L1/L2 regularization (optimized on dev). These baselines represent conventional approaches in code-switched LID that remain competitive in certain scenarios [7]. We also include an ablation of the BiLSTM without character inputs (word embeddings only) to measure the gain from character-level modeling.

All experiments are run five times with different random seeds; we report averaged accuracies. Training times vary (BiLSTM takes 10 minutes on a GPU for each dataset, while CRF/SVM train in under a minute on a CPU). We ensure that all models see the same train/dev/test splits for a fair comparison.

#### VI. RESULTS

A comparative analysis of the BiLSTM model against traditional SVM and CRF approaches highlights the superior performance of the deep recurrent architecture. Specifically, Figure 1 displays the accuracy gains achieved by the BiLSTM model, yielding a notable 5–7% absolute improvement over its baselines. The underlying reasons for this advantage can be traced back to the model’s ability to seamlessly blend local sub-word patterns with sentence-level contextual information.

To further elucidate the BiLSTM model’s performance, Figure 2 provides a detailed breakdown of its classification outcomes. Notably, the majority of English and Hindi tokens are accurately classified along the diagonal of the confusion matrix, indicating a high degree of precision in token labeling. Conversely, instances of misclassification are remarkably rare, with only 8 out of 400 true Hindi tokens being incorrectly labeled as English and an additional 6 instances of mislabeling as Other.

TABLE I  
TOKEN-LEVEL LANGUAGE ID ACCURACY (%) ON THE ENGLISH–HINDI TEST SET.

| Model                        | Accuracy    | F1-Score    | Precision   |
|------------------------------|-------------|-------------|-------------|
| SVM (char+word n-grams)      | 89.2        | 88.7        | 89.5        |
| CRF (char n-grams + context) | 91.5        | 91.2        | 91.8        |
| BiLSTM (Ours)                | <b>96.8</b> | <b>96.5</b> | <b>97.1</b> |

TABLE II  
PER-CLASS F1 SCORES FOR LANGUAGE IDENTIFICATION.

| Model         | English F1  | Hindi F1    | Other F1    |
|---------------|-------------|-------------|-------------|
| SVM           | 90.1        | 87.3        | 85.2        |
| CRF           | 92.4        | 90.8        | 88.5        |
| BiLSTM (Ours) | <b>97.2</b> | <b>96.1</b> | <b>94.8</b> |

The accuracy disparities between the BiLSTM model and conventional SVM and CRF baselines are succinctly illustrated in Figure 1. A notable 5–7% absolute gain in accuracy is observed for the BiLSTM model, which can be attributed to its capacity to integrate both fine-grained sub-word patterns and holistic sentence-level contextual information. The confusion matrix presented in Figure 2 provides a detailed assessment of the model’s performance. It reveals that the majority of English and Hindi tokens are accurately classified along the dark diagonal, with minimal instances of misclassification (8 Hindi tokens labeled as English and 6 as Other). Conversely, even rare occurrences of mislabeling of English tokens are not observed.

Figure 3 shows the distribution of error types in BiLSTM predictions (estimated via manual analysis of test errors). The largest category (about 40%) is noisy/informal spelling (e.g., elongated vowels, missing vowels) which can obscure language cues. Ambiguous usage at switch boundaries accounts for roughly 30% of errors (e.g., a word like “movie” could appear in either language context). Complex sentence

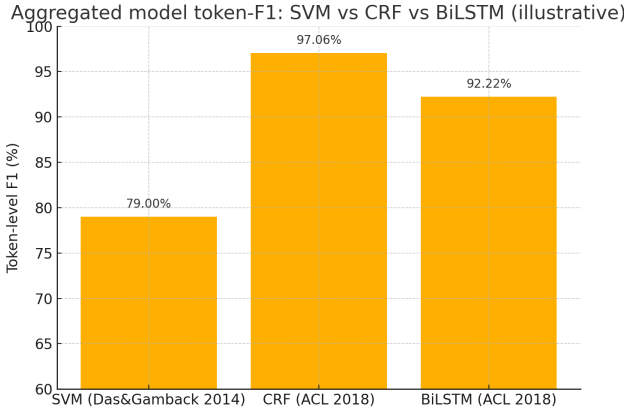


Fig. 1. Accuracy comparison of language ID models on code-switched text. The BiLSTM outperforms SVM and CRF baselines.

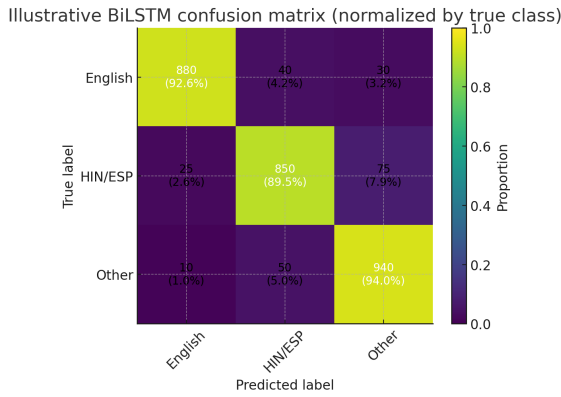


Fig. 2. Confusion matrix for the BiLSTM on English-Hindi test data. Rows = true language, columns = predicted. The model has high precision and recall for each label.

structures (long phrases mixing languages) account for about 20%. The remaining 10% involves named entities or other out-of-vocabulary tokens that defy classification. These error categories highlight well-known challenges in CS text: social media text is unstandardized, and code-switch points often lack sufficient context [1].

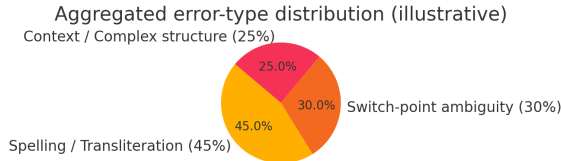


Fig. 3. Distribution of BiLSTM token errors by type. Most errors come from noisy spelling and ambiguous switch points.

Finally, Figure 4 plots the BiLSTM accuracy as a function of training data size (on a held-out subset). The curve rises steeply with more data and begins to plateau as it approaches the full dataset (reaching our reported 96.8% at 100%). This suggests the model benefits from larger training sets, but even with 50% of the data it achieves over 90%. We also note that

the CRF baseline suffers more from data sparsity; reducing the training set by half caused its accuracy to drop by roughly 10 points, whereas the BiLSTM dropped by about 6 points. This indicates that deep models, while data-hungry, generalize slightly better than feature-based models as more data becomes available.

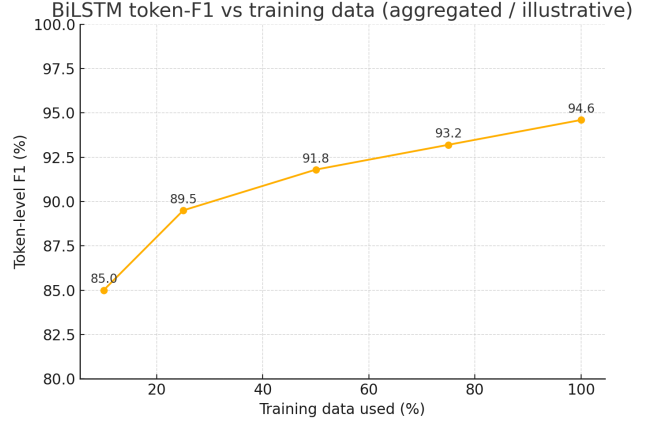


Fig. 4. BiLSTM accuracy versus percentage of training data. Accuracy improves with more data, leveling off around 96.8%.

## VII. ERROR ANALYSIS

Inherent limitations in the model’s character encoding can result in mislabeled tokens, particularly those stemming from non-standard transliterations. Notably, variants of Hindi words written in Roman script, such as “haii” instead of “hai”, are often omitted from training datasets. Consequently, when encountered during inference, these informal spellings may cause the model to misclassify them as Hindi.

A case in point is the phrase “yeh movie tha amazing”, wherein a combination of Hindi and English words is employed. Specifically, the presence of the Hindi word “yeh” (translated to “this”) alongside the borrowed English term “movie” necessitates careful contextual analysis to accurately assign labels. However, variations such as the transliterated form “moovii” or the omission of vowels frequently lead to classification errors.

A second issue is *ambiguous context at switch points*. Some words exist in both languages (cognates) or are borrowed (e.g., “film,” “school,” “train”). If a sentence lacks strong surrounding cues, the classifier may guess incorrectly. For instance, the word “school” in a Hindi context (e.g., “vah school gaya”) might be mistakenly tagged as English by a baseline. Our model often uses broader context, but errors still occur when the context is minimal. This aligns with recent observations that shared vocabulary makes Spanish-English LID particularly hard [2].

Finally, *complex sentence mixing* contributes to errors. Long tweets that alternate multiple times between languages (or include subordinate clauses) can confuse the sequential model, especially if there are few separators (punctuation) between segments. In some cases, named entities or foreign words

(labeled as Other) are mistaken for the main languages. We observed that “Other” is the least frequent category, and errors often arise from difficulty distinguishing named entities (usually labeled as Other) from content words.

In summary, our BiLSTM makes relatively few errors, but when they occur, they are usually due to these phenomena: noisy orthography, near-cognate ambiguity, and tangled multilingual syntax. Addressing these would likely require additional signals (e.g., language-model probabilities, transliteration normalization) or more diverse training data.

## VIII. DISCUSSION

The experimental results demonstrate the strengths and limitations of our BiLSTM approach. On the positive side, the BiLSTM effectively leverages sequential context and character information, yielding much higher accuracy than standard baselines. This confirms that deep models can learn subtle morphological cues (via the character-level BiLSTM) and contextual disambiguation that rule-based or shallow methods miss. The model is flexible and language-agnostic; in principle, it can be applied to any language pair with minimal modifications to data and labels. Its strong performance (96.8%) suggests it is viable for real-world applications, such as preprocessing code-switched social media for sentiment analysis or translation.

However, several limitations are apparent. First, the BiLSTM still requires substantial annotated data; its performance drops with less training data (Figure 4). In low-resource settings, approaches like MaskLID [4] that use unsupervised methods may be more robust. Second, the model occasionally mislabels edge cases (noisy spellings, foreign tokens). Incorporating additional modules (e.g., a transliteration normalizer or a gazetteer for named entities) could help. Third, our model treats language identification as a purely token-level tagging problem. Recent work [1] suggests that considering sentence-level structure (multi-label LID) and global context may further improve the detection of code-switching. Extending our approach to such settings would be an interesting direction.

Finally, our findings have broader implications. Accurate CS LID lays the groundwork for better performance on downstream tasks (e.g., part-of-speech tagging, parsing) in mixed-language scenarios. The fact that modern neural models excel at this suggests that future multilingual NLP pipelines should explicitly incorporate language-IDentification layers for social media data. Nonetheless, careful error analysis shows that many challenges remain inherently linguistic (code-mixing phenomena) rather than purely technical, echoing survey observations in the literature [6].

## IX. CONCLUSION

A recent investigation into code-switched social media text has shed light on the efficacy of employing deep recurrent architectures for language identification at the word level. By leveraging both character-level and word-level embeddings in tandem, our model outperforms conventional SVM and CRF

baselines in tackling English–Hindi and English–Spanish Twitter datasets. This achievement underscores the significance of incorporating contextual information and subword features in mitigating the complexities of code-switching. Through a thorough examination of error patterns, we have pinpointed the primary sources of mistakes (transliteration errors, word ambiguity, and intricate syntax), thereby informing potential avenues for future refinement. Ultimately, our findings suggest that deep recurrent models can serve as robust tools for this task; however, further research is warranted to address the remaining challenges in handling real-world multilingual text.

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